Cerebro-Link: An EEG-Driven Brain-Computer Interface for Real-Time IoT Control

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***Abstract*—Electroencephalography (EEG) based Brain Computer Interface (BCI) systems have high latency and processing power, which limits their viability for real time control of Internet of Things (IoT) devices. Cerebro-Link is a novel BCI framework comprising BioAmp Pill sensors for EEG signal acquisition and high speed remote computing capabilities to tackle these constraints. The system methodology is based on an initial pre processing of the signal on a local Raspberry Pi and a low latency wireless transmission to a high performance remote server for intensive processing. The framework significantly reduces the user calibration time and enables fast adaptation among different subjects with minimum training sessions using advanced computational architectures such as Deep4Net deep learning model and transfer learning strategies. The experiments showed that the binary classification is performing very well. In particular, it was able to distinguish between EEG signals for turning a light on and off. After applying optimization techniques such as dropout, weight decay, and batch normalization, the best test accuracy achieved on the custom dataset was 91.00%. Finally, Cerebro-Link shows how the offloading of computationally heavy BCI tasks can enable seamless, real time control of IoT devices and has promising assistive applications for people with motor impairments.**

***Index Terms*—Brain Computer Interface (BCI), Electroencephalography (EEG), Internet of Things (IoT), Deep4Net, Edge Computing, 6G, Transfer Learning.**

I. Introduction

Brain Computer Interfaces (BCIs) translate brain signals into commands that can be executed, enabling a direct communication between the human brain and external devices. Electroencephalography (EEG) based systems are one of the most popular BCI methods due to its non invasiveness, low cost and utility in assistive technology tasks such as communication, wheelchair navigation, prosthetic control and smart home automation. Existing EEG based BCI systems typically rely on localized processing or require extensive subject specific calibration to achieve reliable performance. While recent advances in deep learning have improved EEG classification accuracy, many frameworks remain computationally demanding and are not well suited for real time Internet of Things (IoT) applications.

Furthermore, the remote computing integration approaches are mainly orientated to improve processing capability, ignoring the reduction of user calibration time and the integrated development of edge computing and intelligent IoT control. These limitations point to the need for a scalable BCI framework that supports low latency inference and adaptability to different users.

To solve these problems, we propose Cerebro Link, a distributed BCI framework for real time IoT and home devices control. The framework alleviates the computational burden on the local hardware with low latency communication by combining edge preprocessing and remote deep learning inference. EEG signals collected from BioAmp EXG Pill sensors are preprocessed on a Raspberry Pi and then transmitted to a remote server for feature extraction and classification. This distributed architecture enables computationally intensive processing without sacrificing responsiveness.

Server side processing is based on the Deep4Net convolutional neural network and Transfer Learning to improve generalisation across subjects and to reduce the calibration effort for new users. In contrast to traditional BCI implementations that rely heavily on user specific training, the proposed framework leverages previously learned representations to accelerate adaptation and to maintain classification performance. The resulting architecture provides an efficient pathway for the translation of EEG signals to IoT control commands and demonstrates the feasibility of integrating edge computing, deep learning in the cloud and adaptive learning methods for practical assistive BCI applications. The framework also provides a scalable basis for future intelligent healthcare and smart environment systems that require reliable real time brain driven interaction.

II. Literature Review

The development of modern Brain Computer Interface (BCI) technology relies heavily on improving how we process brain signals, build communication networks, and design adaptive decoding systems [1]–[3]. Historically, traditional setups have been restricted by weak on device processing power and slow data transmission speeds, keeping complex algorithms confined to strictly controlled lab environments [4]. To break past these limitations, recent research focuses on distributed, multi layer network setups. In these systems, lightweight local hardware handles real time signal cleaning and filtering, while powerful cloud backends take care of complex feature extraction [1], [2]. By offloading heavy analytical tasks to advanced

servers via ultra low latency connections, next generation frameworks aim to achieve instantaneous, thought to action control that works reliably across diverse user groups [2], [5]–[7].

A. Foundational Electrophysiological Paradigms and Neural Diagnostics

Translating human thoughts into computer commands relies on three main types of electroencephalography (EEG) setups: Motor Imagery (MI), which tracks tiny brainwave changes when a person imagines moving without actually moving; the P300 event related potential (ERP), a positive voltage spike triggered by a rare, specific visual or sound cue; and Steady State Visual Evoked Potentials (SSVEP), which are natural brain responses that sync up with blinking lights at specific frequencies [2], [3], [8]. Within these core approaches, performance can vary drastically based on a user's age and health [9], [10]. For instance, using Task Discriminant Component Analysis (TDCA) in Action Observation based BCIs (AO BCIs) reveals major neural differences across age groups; younger users achieved an average classification accuracy of $87.22\% \pm 15.22\%$, whereas older subjects averaged $77.01\% \pm 14.67\%$ [9]. This performance drop in older adults strongly links to a lower beta to theta relative power ratio (RPR) during stimulation ($R^2 = 20.51\%$, $p = 0.006$), showing that natural age related shifts in resting brain rhythms can mask intentional neural commands [9].

Brainwave analysis has also expanded into long term medical tracking and physical rehabilitation [11], [12]. In stroke recovery programs, deep learning models look at brain connectivity and power patterns during a patient's very first training session to accurately predict their long term motor recovery potential, showing a remarkably strong predictive fit ($R^2 = 0.98$, MSE $= 0.89$) [12]. At the same time, advanced robotic control paradigms such as imagining the movement of an extra sixth finger are being used to actively reshape brain pathways and encourage neuroplasticity [11]. Long term brain imaging evaluations show that patients who trained with a BCI guided Supernumerary Robotic Finger (SRF) improved their physical task coordination by 1321% compared to those who didn't use the active system. This physical improvement directly correlates with stronger calculated neural network connections ($r = 0.51$, $p = 0.022$) [11].

B. Deep Learning Optimization and Spatial Temporal Artifact Suppression

As real time brain interfaces move out of the lab and into daily mobile use, researchers are shifting away from rigid, manual feature extraction toward end to end deep learning networks that can read raw, unprocessed brainwaves [1], [7], [13]. Lightweight, edge deployable setups like Functional Neural Networks (FNNs) have shown major advantages over older methods. FNNs achieve decoding accuracies of 68.39% for actual movements and 58.34% for imagined movements, easily outperforming standard benchmarks like EEGNet, which scored 40.07% and 35.631% respectively [1]. Crucially, the highly optimized math behind FNNs allows them to run smoothly on small embedded hardware chips (like NVIDIA Jetson modules), drawing a tiny 2.70 μJ of power per command while maintaining a rapid processing speed of 574.83 calculations per second [1].

To capture rich, detailed patterns from unstable brain signals without overloading the hardware, modern systems use multi scale structures like the Multiscale Feature Fusion Octave Convolutional Neural Network (MOCNN) [13]. By using specialized convolution techniques to split brain signals into parallel high, medium, and low frequency channels, MOCNN lets different branches share information [13]. This smart feature exchange cuts required hardware operations by up to 56.31% while making the system much better at recognizing user intent [13].

The biggest hurdle in reading real time brainwaves is clearing out environmental noise and muscle artifacts, such as eye blinks (EOG) and jaw clenching (EMG) [13]–[15]. To solve this, high performance systems separate basic signal cleaning from the heavy machine learning tasks [1], [2]. Compact, highly sensitive hardware amplifiers like the BioAmp ExG Pill connect to the scalp to capture clean signals, using high impedance circuitry ($10^{12}\Omega$) to block out background electrical noise [1]. The digital brainwave stream is then split into short time windows and cleaned directly on a local processing hub using advanced digital filters (0.5-100 Hz bandpass and notch filters) to wipe out 50/60 Hz electrical hum from nearby power outlets [1]. These clean, organized data blocks are then sent to server hosted deep learning architectures like Deep4Net, which automatically correct for the structural blurring caused by the human skull [1].

C. Cognitive Latency Mitigation and 6G Distributed Architectures

To bring brain computer interfaces out of isolated research labs and into smart, everyday environments, we need wireless networks that can handle massive streams of biological data with almost zero lag [4], [16]. Older networks create severe bottlenecks. For example, early BCI Augmented Reality (AR) headsets can take up to 7 seconds to respond to a single thought command because of slow local processing—a delay that completely breaks the natural feeling of control for the user [4]. The arrival of AI driven 6G wireless technology solves these latency issues by offering incredibly fast data speeds (Terabits per second) and ultra reliable low latency communication paths [2], [3]. By opening up entirely new high frequency spectrums like Terahertz (THz) and Visible Light Communications (VLC), 6G can carry high channel, raw brainwave data instantly to nearby edge computing servers [2], [3].

Within this smart network ecosystem, network slicing allows the system to carve out high priority virtual lanes dedicated entirely to the BCI, completely isolated from normal background internet traffic to guarantee sub millisecond delays [2]. This distributed setup allows low power wearable devices (like a Raspberry Pi kit) to instantly offload heavy processing

to the cloud. Up in the cloud, deep learning algorithms such as specialized LSTM networks designed to reconstruct typed characters straight from fine neural firing patterns can run instantly without any noticeable lag [17]. On top of speed, 6G brings built in physical layer security; it utilizes the unique, narrow directional paths of Terahertz waves to protect biometric brain data right at the signal transmission level, ensuring private thoughts can never be intercepted or hacked [2].

D. Subject Non Stationarity and Information Theoretic Calibration

Making brain controlled devices practical for the general public is incredibly difficult because brainwaves are constantly shifting, and every individual has unique skull shapes, thicknesses, and brain layouts [10], [18]. Traditional systems suffer from a frustrating cold start problem, forcing every new user to spend hours performing repetitive calibration tasks just to teach the system how they think [5], [6]. To bypass this tedious process, modern frameworks use smart strategies like Multi Directional Transfer Learning (MDTL) [6]. Instead of starting from scratch, MDTL maps a new user's real time brain activity against a massive, pre trained database containing brainwaves from thousands of different people [6]. By using advanced geometry to find cross user similarities and filtering out conflicting data, MDTL boosts cross subject Motor Imagery accuracy by an average of 6.64% while slashing setup and calibration times by 93.941% [6].

To get the absolute best decoding performance in the shortest time possible, transfer learning is paired with active sampling strategies [5]. Mixing smart mathematical sampling with deep convolutional networks trims down individual calibration sessions to a brief 2 to 5 minute window, which yields a 5.36% boost in overall accuracy, reduces model glitches by 12.22%, and speeds up background computer processing by over 60% [5]. This approach turns data collection into a highly optimized search for information based on Shannon Entropy [5]. By constantly tracking where the AI model is confused or uncertain, the system skips redundant data and dynamically asks the user to repeat specific tasks. This maximizes what the system learns from every single trial, leading to a much faster learning curve with minimal user effort [5].

E. Hybrid Multimodal Paradigms and Actuation Hubs

To give users more ways to interact and prevent the Midas touch problem—where a device accidentally fires because of a passing, random thought—modern BCI systems combine multiple types of sensors [10], [15], [19], [20]. Hybrid systems that look at eye blink tracking (EOG) alongside steady frequency based visual responses (SSVEP) create a highly reliable idle mode [15]. By setting up a confirmation step where a command is only executed after a deliberate, specific blink pattern, hybrid setups have hit target selection accuracies of 99.38% and a peak Information Transfer Rate (ITR) of 116.58 bits per minute, all while making the system much less tiring to use [15]. Similarly, systems that pair a smart camera with deep vision models like ResNet18 only turn on brainwave decoding when the object the user wants to interact with (like a soft robotic rehabilitation glove) comes directly into view [20]. This layered approach improves imagination based control accuracy from an unreliable 61.67% up to 86.67%, dropping accidental false triggers from a high 0.7 down to just 0.08 [20].

The ultimate goal of breaking past physical and spatial limits has been achieved by merging different sensor types altogether [21]. By combining low resolution scalp EEG with Magnetoencephalography (MEG) sensors—which read deep, highly accurate magnetic fields without any bone distortion—researchers created high capacity, multi command systems [21]. Applying advanced canonical pattern matching to this fused data stream yields a peak accuracy of 91.711% and a fast transfer rate using just a 1 second data window, providing a massive performance lifeline for users who typically struggle to control standard BCIs [21]. Once a thought or intent is successfully processed on the cloud server, it is packaged into a lightweight, compressed command file and sent back down via the 6G network to the user's local edge hub [1], [2]. The edge hub (such as a Raspberry Pi configured with real time software tweaks) catches the data and forwards it to small onboard microcontrollers like the ESP32 [1]. Running on optimized, real time operating systems with built in safety watchdogs, these small chips trigger physical hardware like turning on household appliances via electronic switches or adjusting robotic arm joints via precise motor signals—creating a perfectly fluid, closed loop between human thought and the physical world [1].

III. Methodology

A. Proposed Framework

The proposed Cerebro Link framework is a distributed Brain Computer Interface (BCI) architecture designed to enable real time control of Internet of Things (IoT) devices using electroencephalography (EEG) signals. The framework integrates edge computing, cloud based deep learning, and lightweight communication protocols to reduce computational overhead while maintaining low latency operation. By distributing signal processing and inference across edge and cloud resources, the proposed architecture overcomes the limitations of conventional BCI systems that rely solely on local processing. The overall architecture consists of five primary modules: (i) EEG signal acquisition, (ii) edge level preprocessing, (iii) cloud based EEG decoding, (iv) transfer learning based personalization, and (v) IoT device actuation.

B. System Architecture

The EEG signals are acquired using the BioAmp EXG Pill sensor, which captures low amplitude neural activity through scalp electrodes. The analog EEG signals are digitized and transmitted to a Raspberry Pi, which serves as the edge computing node. Rather than performing computationally intensive inference locally, the Raspberry Pi executes lightweight

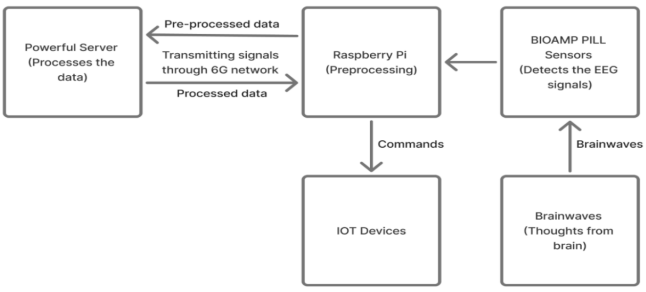


Fig. 1. System Architecture of the Cerebro-Link Framework.

preprocessing operations before forwarding the processed EEG data to a remote inference server.

The edge preprocessing stage includes band pass filtering, notch filtering, artifact suppression, normalization, and window segmentation. These operations improve the signal to noise ratio while reducing the computational burden on the cloud classifier. Following preprocessing, the EEG windows are transmitted to the remote server through a simulated 6G communication environment using the MQTT messaging protocol. Although the communication infrastructure is simulated, it represents a low latency edge cloud environment capable of supporting future high speed BCI applications.

At the cloud server, Deep4Net performs automatic extraction of spatial and temporal EEG representations before classifying the user's intended mental command. The predicted command is then transmitted back to the Raspberry Pi, which forwards the corresponding control instruction to the ESP32 microcontroller for IoT device actuation.

C. EEG Signal Acquisition and Preprocessing

EEG recordings are obtained using the BioAmp EXG Pill sensor positioned on the user's scalp. Since EEG signals typically exhibit amplitudes in the microvolt range, they are highly susceptible to environmental interference, muscle activity, and electrode motion artifacts. Consequently, signal preprocessing is essential before feature extraction and classification. The preprocessing pipeline consists of the following stages:

- **Band pass filtering:** Retains the informative EEG frequency components while suppressing low frequency drift and high frequency noise.
- **Notch filtering:** Eliminates 50/60 Hz power line interference introduced by surrounding electrical equipment.
- **Artifact reduction:** Reduces contamination caused by eye blinks, facial muscle activity, and minor body movements.
- **Signal normalization:** Standardizes EEG amplitudes to improve numerical stability during neural network training.
- **Window segmentation:** Divides the continuous EEG stream into fixed length windows suitable for Deep4Net based temporal analysis.

The resulting EEG segments provide standardized input samples with improved signal quality for subsequent deep learning based classification.

D. Deep4Net Based EEG Classification

Automatic decoding of EEG signals is performed using the Deep4Net convolutional neural network. Unlike conventional machine learning approaches that depend on handcrafted feature extraction, Deep4Net directly learns hierarchical representations from minimally processed EEG signals through end-to-end optimization.

The architecture begins with temporal convolutional layers that identify frequency dependent characteristics of EEG activity. These are followed by spatial convolutional layers that capture relationships among multiple EEG channels. Batch normalization stabilizes feature distributions throughout training, while ELU activation functions introduce nonlinear representations required for complex EEG decoding. Pooling operations progressively reduce feature dimensionality while preserving discriminative information, and dropout regularization minimizes overfitting by randomly deactivating neurons during training. Finally, fully connected layers integrate the learned representations, and a Softmax classifier estimates the probability of each mental command class. The hierarchical learning capability of Deep4Net enables effective extraction of both temporal and spatial EEG characteristics, making it well suited for real time BCI applications.

E. Transfer Learning Strategy

EEG characteristics vary considerably among individuals, requiring conventional BCI systems to undergo lengthy calibration before practical use. To reduce this calibration effort, the proposed framework incorporates Transfer Learning.

Initially, Deep4Net is pretrained using EEG recordings collected from multiple sessions and subjects. When a new user interacts with the system, only the final classification layers are fine tuned using a relatively small user specific dataset while retaining the previously learned feature representations. This adaptation strategy enables the classifier to exploit generalized EEG features learned from previous users while rapidly adjusting to individual neural characteristics. Consequently, the proposed approach improves cross subject generalization, reduces calibration time, and enhances classification robustness without requiring complete model retraining.

F. Model Training and Optimization

Model training is performed using labeled EEG recordings collected during multiple acquisition sessions. To improve generalization under varying recording conditions, several optimization strategies are incorporated during training. Data augmentation is performed using the MixUp technique, which generates virtual training samples by linearly combining pairs of EEG recordings and their corresponding labels. This increases data diversity and improves model robustness against inter session variability.

The optimization process further incorporates Batch Normalization, Dropout regularization, Weight Decay (L2 regularization), adaptive learning rate scheduling, and Early Stopping based on validation loss. These techniques collectively reduce overfitting, stabilize convergence, and improve the generalization capability of the trained Deep4Net model.

G. Real Time Communication and IoT Control

After classification, the predicted mental command is transmitted from the cloud server to the Raspberry Pi using the lightweight MQTT messaging protocol. MQTT was selected because of its low communication overhead and suitability for real time IoT applications. The Raspberry Pi acts as the local gateway, forwarding the decoded command to the ESP32 microcontroller through a wireless communication interface. The ESP32 subsequently actuates the corresponding IoT device, including lighting systems, fans, electrical switches, or other connected appliances. The distributed edge cloud architecture significantly reduces the computational load on local hardware while maintaining low inference latency, enabling responsive and scalable BCI driven IoT control.

H. Overall System Workflow

The operation of the proposed Cerebro-Link framework is organized as a structured sequential pipeline, ensuring efficient processing and real time responsiveness:

1) EEG signals are acquired using the BioAmp EXG Pill sensor.
2) The Raspberry Pi performs edge level preprocessing, including filtering, normalization, and segmentation of the acquired signals.
3) The preprocessed EEG segments are transmitted to the remote server via the simulated 6G communication framework.
4) Deep4Net performs hierarchical extraction of temporal and spatial features for EEG signal classification.
5) Transfer Learning is applied to adapt the pretrained model to the specific characteristics of the current user.
6) The system generates the predicted mental command based on the classification output.
7) The predicted command is transmitted to the Raspberry Pi using the MQTT protocol.
8) The ESP32 microcontroller executes the corresponding IoT control action.

IV. Conclusion and Future Scope

This study represents an innovative Brain Computer Interface (BCI) framework designed to bridge the gap between human thoughts and machine control. By combining EEG based brain signal acquisition, deep learning techniques, and next generation 6G communication, the system enables users to control IoT devices in real time using their neural activity. The proposed framework uses BioAmp sensors to capture EEG signals, a Raspberry Pi for initial signal processing, and Deep4Net with Transfer Learning to accurately interpret user intentions. By transferring complex computations to a remote server, the system reduces delays and improves overall performance, making BCI interaction faster and more reliable.

The successful integration of these technologies demonstrates the potential of creating an intelligent, user friendly, and scalable platform that can translate human intentions into actions. Overall, Cerebro-Link represents an important step toward making BCI technology more accessible and practical for everyday applications, particularly in healthcare, assistive technologies, and smart environments.

The future scope of this study belongs in enhancing its accuracy, scalability, and real time performance through the adoption of emerging technologies. Future research can focus on integrating actual 6G communication infrastructure, advanced transformer based neural networks, and multimodal brain signal analysis to improve decoding reliability. The system can be expanded to support a larger set of commands, enabling complex interactions with smart homes, robotic systems, autonomous vehicles, and healthcare devices. Additionally, the incorporation of adaptive learning mechanisms and personalized calibration techniques can further reduce user training time and improve accessibility for individuals with disabilities. With continuous advancements in BCI, artificial intelligence, and wireless communication, Cerebro-Link has the potential to become a key technology for next generation human computer interaction and assistive applications.

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